Project Report

on

SafeRoute - AI That Understands Safety Beyond Maps

Subject: Machine Learning Lab

Subject Code: 25CAP-672

Submitted By

Priyanka Chandwani 25MCI10122

Under the guidance of

Mrs. Deeksha Baweja

In the partial fulfilment for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING(AI&ML)

Chandigarh University

May 2026

CERTIFICATE

This is to certify that Priyanka Chandwani, a student of Master of Computer Application – Artificial Intelligence and Machine Learning, has successfully completed the Minor Project titled "SafeRoute – AI That Understands Safety Beyond Maps" under the esteemed guidance of Mrs. Deeksha Baweja, Assistant Professor, University Institute of Computing (UIC), Chandigarh University.

This project was undertaken as part of the academic curriculum and is submitted in partial fulfilment of the requirements for the MCA programme. The work presented in this project is a result of independent research, diligent effort, and dedication, demonstrating the student's ability to apply theoretical knowledge to practical problem-solving in the field of Machine Learning.

The project, "SafeRoute – AI That Understands Safety Beyond Maps", is a web-based intelligent system that recommends the safest travel routes by analyzing multiple factors such as crime data, lighting conditions, crowd density, and social sentiment. The system leverages Machine Learning techniques such as clustering for hotspot detection and classification for safety prediction, along with an innovative Emotional Safety Score to provide a more comprehensive measure of safety. It integrates a Python-based backend for data processing and prediction with a modern web-based frontend for seamless user interaction.

I hereby confirm that this project is an original work carried out by the student.

Project Guide:
Mrs. Deeksha Baweja
Assistant Professor
University Institute of Computing
Chandigarh University

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Chandigarh University and the University Institute of Computing (UIC) for providing me with the opportunity, resources, and infrastructure to undertake this project, **"SafeRoute – AI That Understands Safety Beyond Maps"**.

I extend my heartfelt appreciation to my respected mentor, Mrs. Deeksha Baweja, Assistant Professor, for her invaluable guidance, continuous encouragement, and insightful feedback throughout the development of this project. Her expertise in Machine Learning and Artificial Intelligence played a vital role in the successful completion of this work.

I am also deeply thankful to the open-source communities, including **scikit-learn, pandas, NumPy, and Flask**, for providing powerful, well-documented tools and libraries that made the development of this project efficient and effective.
I am grateful to my friends and peers for their constant motivation, constructive suggestions, and support during the project's development. Their collaboration helped me refine my understanding and approach to building AI-powered applications.

Lastly, I express my profound gratitude to my family for their unwavering support, patience, and encouragement throughout this learning journey. This project has been an enriching experience, enhancing my understanding of machine learning techniques such as clustering and classification, data analysis, predictive modeling, and full-stack system integration.

Priyanka Chandwani
MCA – Artificial Intelligence and Machine Learning
Chandigarh University

ABSTRACT

With the rapid growth of urbanization and increasing concerns about personal safety, especially in unfamiliar areas, the need for intelligent navigation systems that go beyond shortest-path recommendations has become crucial. Traditional navigation systems primarily focus on distance and time, often neglecting critical safety factors such as crime rates, environmental conditions, and public sentiment.

SafeRoute – AI That Understands Safety Beyond Maps is an AI-powered intelligent navigation system developed to address this challenge. The system recommends the safest travel routes by analyzing multiple parameters including crime data, lighting conditions, crowd density, and social sentiment. It introduces an innovative concept of an **Emotional Safety Score**, which reflects how safe a location feels based on real-world data and public perception.

The machine learning component is implemented using Python and leverages libraries such as **scikit-learn, pandas, and NumPy**. Clustering algorithms like K-Means are used to identify high-risk zones or crime hotspots, while classification models are employed to predict the safety level of different routes. Data preprocessing techniques, feature engineering, and model evaluation metrics are applied to ensure accurate and reliable predictions.

The system is designed with a modular architecture, where the backend handles data processing, machine learning predictions, and safety scoring, while the frontend provides an intuitive interface for users to input source and destination points and receive optimized safe routes. The integration ensures seamless communication between components, delivering real-time recommendations.
This project successfully demonstrates the practical application of machine learning, data analysis, and intelligent decision-making in a real-world scenario. It covers core concepts such as clustering, classification, predictive modeling, and full-stack integration, making it a comprehensive solution aligned with the objectives of the Machine Learning curriculum.

CHANDIGARH UNIVERSITY
Discover. Learn. Empower.

NAAC GRADE A+
Accredited University

CONTENTS

1. INTRODUCTION

With the increasing pace of urbanization and the growing reliance on digital navigation systems, ensuring personal safety during travel has become a significant concern. Most existing navigation applications focus primarily on optimizing routes based on distance and time, often overlooking critical safety-related factors such as crime levels, environmental conditions, and public sentiment. This gap highlights the need for an intelligent system that prioritizes safety alongside efficiency.

SafeRoute – AI That Understands Safety Beyond Maps is designed to address this challenge by integrating machine learning techniques with real-world data to recommend safer travel routes. The system analyzes multiple parameters including crime rate, lighting conditions, crowd density, and social sentiment to evaluate the safety of different locations and routes.

To achieve this, **K-Means clustering** is used to group locations based on features such as crime rate, latitude, and longitude. This helps in identifying crime hotspots within a city. The centroids of these clusters represent the average characteristics of each region and are further utilized to categorize areas into high, medium, or low crime zones.

In addition, **Logistic Regression** is employed as a classification technique to predict the safety level of a given location. The model takes into account various features such as crime rate, lighting conditions, crowd density, and sentiment score. By learning the importance (weights) of each feature, the model determines the probability of a location being safe or unsafe. Based on this probability, new locations are classified accordingly.

The integration of clustering and classification enables the system to not only identify unsafe zones but also make intelligent predictions about safety in real-time. This approach enhances traditional navigation by incorporating both physical and emotional aspects of safety, ultimately providing users with more reliable and secure route recommendations.

2. OBJECTIVES

- To develop an intelligent navigation system that prioritizes user safety along with route efficiency.
- To analyze multiple real-world factors such as crime rate, lighting conditions, crowd density, and social sentiment for evaluating location safety.
- To implement K-Means clustering for identifying and visualizing crime hotspots based on geographical and crime-related data.
- To build a Logistic Regression model to classify locations as safe or unsafe based on various influencing features.
- To introduce an Emotional Safety Score that reflects both actual and perceived safety of a location.
- To design a system that provides data-driven safe route recommendations to users.
- To integrate machine learning models with a user-friendly web interface for real-time interaction and decision-making.
- To demonstrate the practical application of machine learning techniques such as clustering, classification, and predictive modeling in a real-world problem.

3. TOOLS AND LIBRARIES USED

Programming Language:

- Python – Used for implementing machine learning models, data processing, and backend logic.

Libraries:

- NumPy – For numerical computations and handling arrays efficiently.
- pandas – For data manipulation, preprocessing, and analysis.
- scikit-learn – For implementing machine learning algorithms such as K-Means clustering and Logistic Regression.

Machine Learning Techniques

- **K-Means Clustering** – To identify crime hotspots based on location and crime data.
- **Logistic Regression** – To classify locations as safe or unsafe.

API Framework:

- Flask (Python Framework) – Used to build REST APIs for handling requests and serving ML model predictions.

Development Tools:

- VS Code – The primary code editor used for writing and running the Python ML code and Flask API.

Dataset & Data Sources

- Crime datasets (real or simulated)
- Location data (latitude & longitude)
- Sentiment data (from social media or sample dataset)

4. IMPLEMENTATION

Code: app.py

```
from flask import Flask, render_template, request, jsonify
from model import SafetyModel
from utils import compute_ess, find_route, haversine_km
from india_data import get_all_locations, get_cities, search_locations

app = Flask(__name__)

model = SafetyModel()
model.train()
@app.route("/")
def home():
    return render_template("index.html")
@app.route("/api/search")
def api_search():
    q = request.args.get("q", "").strip()
    if len(q) < 2:
        return jsonify([])
    matches = search_locations(q)[:8]
    return jsonify([
        {"label": f"{m['name']}, {m['city']}", "name": m["name"],
         "city": m["city"], "state": m["state"],
         "lat": m["lat"], "lng": m["lng"]}
        for m in matches
    ])
@app.route("/api/cities")
def api_cities():
    return jsonify(get_cities())
@app.route("/predict", methods=["POST"])
def predict():
    data = request.get_json()
    source  = data.get("source", "Connaught Place")
    dest    = data.get("destination", "Saket")
    time_of_day = data.get("time", "day")
    route, all_locs, graph, src_idx, dst_idx, light = \
        find_route(source, dest, time_of_day)
    enriched = []
    for wp in route:
        pred = model.predict(wp["crime_rate"], wp["lighting"],
                             wp["crowd_density"], wp["sentiment"])
```

```
        enriched.append({**wp, "prediction": pred})

    avg_ess = sum(w["ess"] for w in enriched) / max(len(enriched), 1)
    overall = "safe" if avg_ess >= 0.5 else "unsafe"
    import heapq, math, random
    def alt_dijkstra(noise=0.3):
        dist_map = {n: math.inf for n in graph}
        dist_map[src_idx] = 0.0
        prev = {n: None for n in graph}
        pq = [(0.0, src_idx)]
        vis = set()
        while pq:
            cost, u = heapq.heappop(pq)
            if u in vis: continue
            vis.add(u)
            if u == dst_idx: break
            for v, w, _ in graph.get(u, []):
                nc = cost + w * random.uniform(1 - noise, 1 + noise)
                if nc < dist_map[v]:
                    dist_map[v] = nc
                    prev[v] = u
                    heapq.heappush(pq, (nc, v))
        path, cur = [], dst_idx
        while cur is not None:
            path.append(cur)
            cur = prev[cur]
        path.reverse()
        return path if path[0] == src_idx else [src_idx, dst_idx]

    alt_routes = []
    seen_paths = {tuple(w["name"] for w in enriched)}
    random.seed(42)
    for _ in range(6):
        ap = alt_dijkstra(noise=0.4)
        ap_wps = []
        for idx in ap:
            l = all_locs[idx]
            ess = compute_ess(l["crime_rate"], light, l["crowd_density"],
l["sentiment"])
            ap_wps.append({"name": l["name"], "city": l["city"],
                           "lat": l["lat"], "lng": l["lng"], "ess": ess})
        key = tuple(w["name"] for w in ap_wps)
```

```
            if key not in seen_paths:
                seen_paths.add(key)
                avg = sum(w["ess"] for w in ap_wps) / max(len(ap_wps), 1)
                alt_routes.append({"path": ap_wps, "avg_ess": round(avg, 3)})
            if len(alt_routes) >= 2:
                break

        # breakdown averages
        avg_crime  = sum(w["crime_rate"] for w in enriched) / len(enriched)
        avg_crowd  = sum(w["crowd_density"] for w in enriched) / len(enriched)
        avg_sent   = sum(w["sentiment"] for w in enriched) / len(enriched)

        return jsonify({
            "source": source, "destination": dest, "time": time_of_day,
            "route": enriched,
            "avg_ess": round(avg_ess, 3),
            "overall_prediction": overall,
            "breakdown": {
                "crime_safety": round(1 - avg_crime, 3),
                "lighting": light,
                "crowd_density": round(avg_crowd, 3),
                "sentiment": round(avg_sent, 3),
            },
            "alternative_routes": alt_routes,
            "total_waypoints": len(enriched),
        })

    @app.route("/api/heatmap")
    def api_heatmap():
        return jsonify({"hotspots": model.hotspots(), "total": len(model.hotspots())})

    @app.route("/api/all_locations")
    def api_all_locations():
        locs = get_all_locations()
        return jsonify([{
            "name": l["name"], "city": l["city"], "state": l["state"],
            "lat": l["lat"], "lng": l["lng"],
            "ess": l["ess"], "crime_rate": l["crime_rate"],
            "safety_label": l["safety_label"]
        } for l in locs])
    if __name__ == "__main__":
        app.run(debug=True, port=5000)
```

5. RESULT(Screenshot)

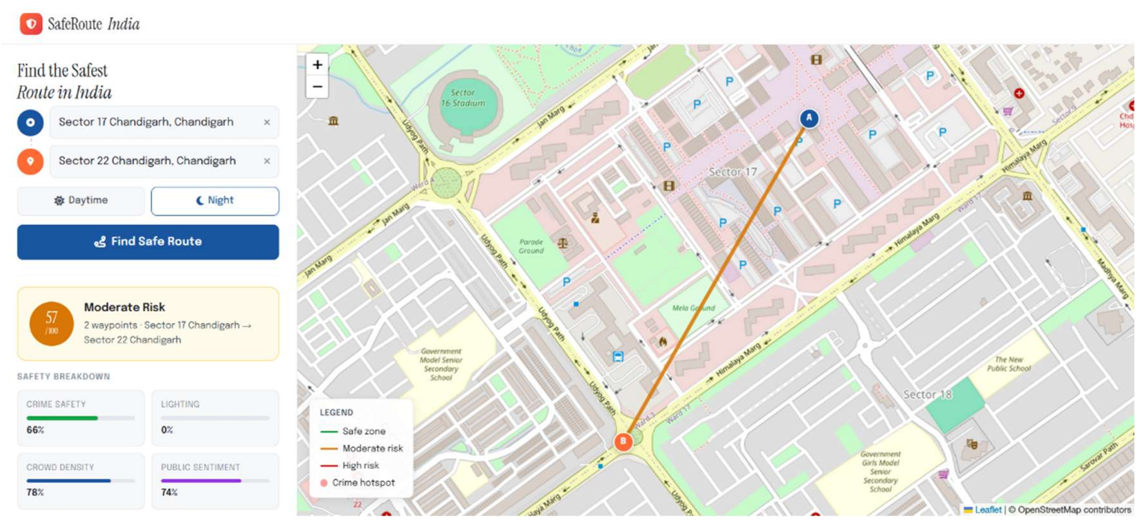

Figure 1: Saferoute between 2 distances(Moderate Risk)

6. CONCLUSION

The SafeRoute – AI That Understands Safety Beyond Maps project successfully demonstrates how machine learning can be applied to enhance traditional navigation systems by incorporating safety as a key factor. Unlike conventional route recommendation systems that focus only on distance and time, this project introduces a more holistic approach by considering multiple real-world parameters such as crime rate, lighting conditions, crowd density, and social sentiment.

The use of K-Means clustering effectively identifies crime-prone areas and helps in visualizing hotspots, while Logistic Regression enables accurate classification of locations as safe or unsafe. The integration of these techniques, along with the innovative concept of an Emotional Safety Score, allows the system to provide more reliable and user-centric route recommendations.

This project not only highlights the practical implementation of machine learning concepts such as clustering, classification, and predictive modeling but also demonstrates the importance of combining data-driven insights with real-world applications. The successful integration of backend processing and frontend interaction ensures a seamless user experience.

In conclusion, SafeRoute serves as a step forward in building smarter and safer navigation systems. It has the potential to be further enhanced by incorporating real-time data, advanced AI models, and large-scale deployment, making it highly relevant for smart cities and future intelligent transportation systems.

7. REFERENCES

- GitHub Repository: https://github.com/priyanka7304/ml-project.git
- Scikit-learn: Machine Learning in Python. Available at: https://scikitlearn.org/stable/
- Flask Documentation – Pallets Projects. Available at: https://flask.palletsprojects.com/
- Pandas Documentation. Available at: https://pandas.pydata.org/docs/
- NumPy Documentation. Available at: https://numpy.org/doc/

Open datasets for crime and location analysis:

- https://data.gov.in
- https://www.kaggle.com